

Yacin Ben Kacem

Final-Year Software Engineering Student — Agentic AI Systems & Secure Backend Infrastructure
yacinbenkacem19@gmail.com | yacinbenkacem.me | +216 29 504 806 | linkedin.com/in/yacin-ben-kacem | github.com/YACINBK

SUMMARY

Final-year Software Engineering student at ENISO focused on agentic AI, secure backend orchestration, and automation pipelines built for real integration constraints. Connects LLMs, RAG, OCR, and tool-calling agents to production-minded backends with identity, caching, containerization, and fallback-aware delivery.

EXPERIENCE

White Cape Technologies — AI & Automation Intern Sousse, Tunisia
Agentic RAG Knowledge Assistant Jun – Aug 2026

- Built an internal AI assistant so engineering teams can query company knowledge and codebases in plain language instead of manual search — architected as a **9-node LangGraph state machine** (classify → retrieve → rerank → relevance gate → generate → faithfulness check) returning grounded, cited, role-aware answers streamed live over SSE.
- Ran a **fully local inference stack** (LiteLLM → Ollama, Qdrant, BGE-M3 embeddings, bge-reranker-v2-m3 via Hugging Face TEI), keeping all proprietary document content on-network for compliance.
- Enforced role- and admin-scoped access **inside the retrieval query itself**, not the UI, behind Keycloak OIDC session auth with CSRF protection — so users only ever see content they're authorized for.
- Gated every response on a faithfulness check that declines rather than hallucinates, backed by Celery ingestion, Qdrant semantic caching, and **218 contract-driven tests** — keeping answers reliable enough for engineers to act on.

UX Insight Platform Jun – Aug 2025

- Delivered an **AI-driven UX-audit platform** that turns screenshots, OCR, interaction telemetry, and visual heuristics into structured, actionable findings for product teams (Angular · Spring Boot gateway · FastAPI services).
- Moved heavy AI work off the request path into an **async job lifecycle** with queueing, polling, and recovery; secured the gateway with Keycloak OAuth2/OIDC and JWT validation.
- Isolated LLM-inference, OCR, and image-processing workloads into fallback-aware FastAPI microservices on PostgreSQL, Redis caching, and Docker Compose.

PROJECTS

QuickFlow — AI Productivity Assistant 2025

- Built a full-stack AI productivity assistant for meeting summaries, structured minutes, and professional email generation.
- Developed Spring Boot backend services and Angular frontend workflows; integrated secure identity management with Keycloak and social login support.
- Designed reusable communication automation flows, keeping authentication and workflow reliability as first-class concerns.

AI Image Vectorization Pipeline 2025

- Designed a Python AI image-generation and vectorization pipeline for apparel design workflows under **8GB VRAM constraints**.
- Automated preprocessing, SAM segmentation, Oklab color simplification, and SVG tracing into repeatable vector-ready outputs.
- Treated hardware limits and repeatability as design constraints, with VRAM flushing and batch checkpointing for recovery.

Agentic Video Production Pipeline 2026

- Built an agentic video-production pipeline that runs **unattended end-to-end**, inspecting filesystem state before each step to skip already-completed work instead of reprocessing from scratch.
- Designed a **self-correcting generation loop** that detects rejected or malformed outputs and automatically reformulates the prompt and retries, rather than failing the run.
- Hardened browser-automation steps with **Playwright selector fallbacks and retry logic**, and made the pipeline resumable after interruption — picking up mid-run instead of restarting.

SKILLS

Programming: Python, Java, C/C++, C#, TypeScript/JavaScript, SQL
Agentic AI & LLMs: LangGraph, LangChain, LiteLLM, local LLMs (Ollama), RAG pipelines, reranking (TEI), semantic caching, faithfulness checks, tool-calling agents, MCP servers, prompt engineering
Machine Learning & Data: NumPy, pandas, Pydantic, scikit-learn, PyTorch, Keras, LightGBM, data preprocessing, feature engineering
Backend & Security: Spring Boot, FastAPI, Flask, Keycloak, OAuth2/OIDC, JWT, PostgreSQL, Redis, Qdrant, Celery, SSE
Automation, Vision & Media: Playwright browser automation, selector fallbacks, retry logic, queue orchestration, Segment Anything (SAM), VTracer/SVG tracing, Oklab quantization, OpenCV/Pillow, FFmpeg/Whisper
Frontend & Delivery: Angular, React, HTMX, Streamlit, Gradio, Docker & Compose, Linux CLI, Git/GitHub/GitLab, Postman, caching, fallback-aware workflows

CERTIFICATIONS

NVIDIA DLI: [Prompt Engineering for LLMs](#) · [RAG Agents with LLMs](#) · [AI Cybersecurity Pipelines](#) · [Transformer NLP](#) · [Diffusion Models](#) · [Real-Time Video AI](#) · [Predictive Maintenance](#) · [Fundamentals of Deep Learning](#)
DeepLearning.AI: [Neural Networks and Deep Learning](#) · [Supervised Machine Learning](#) | **Google Cloud:** [Introduction to Generative AI](#)
Competition: [IndabaX Tunisia 2025 Challenge 1](#) — top 50% (31 / 69), 20-hour applied-ML hackathon on 4G/5G telecom KPI anomaly detection

EDUCATION

École Nationale d'Ingénieurs de Sousse (ENISO) Sep 2024 – Present
Engineering Degree in Software Engineering
Preparatory Institute for Engineering Studies of Monastir Sep 2022 – Jun 2024
Mathematics & Physics Preparatory Cycle · National Rank: 509 / 1800

LANGUAGES

Arabic: Native | **English:** C1 | **French:** B2 | **German:** A1